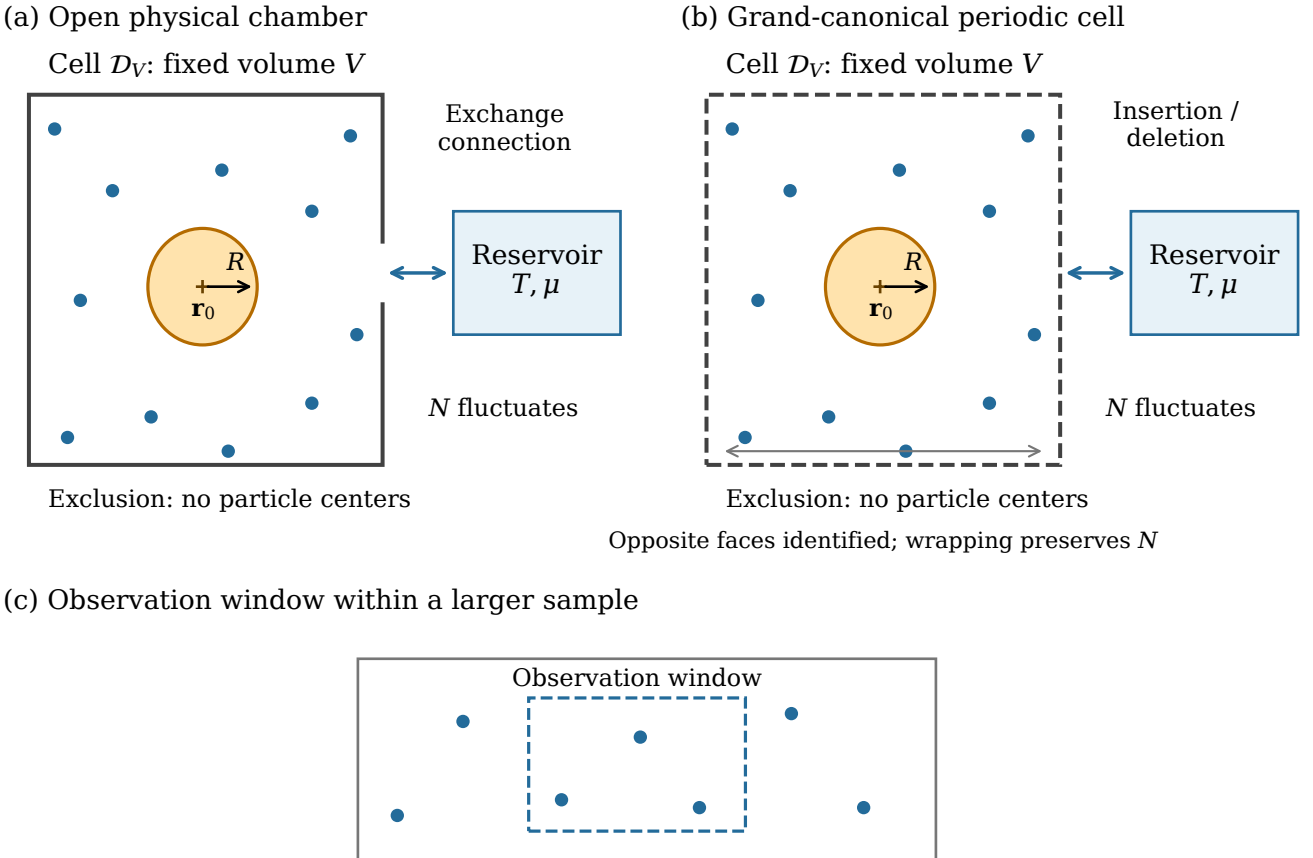
(a) Open physical chamber
Cell $\mathcal{D}_V$: fixed volume V
Exchange connection
Reservoir T, μ
R
$\mathbf{r}_0$
N fluctuates
Exclusion: no particle centers
(b) Grand-canonical periodic cell
Cell $\mathcal{D}_V$: fixed volume V
Insertion / deletion
Reservoir T, μ
R
$\mathbf{r}_0$
N fluctuates
Exclusion: no particle centers
Opposite faces identified; wrapping preserves N
(c) Observation window within a larger sample
Observation window